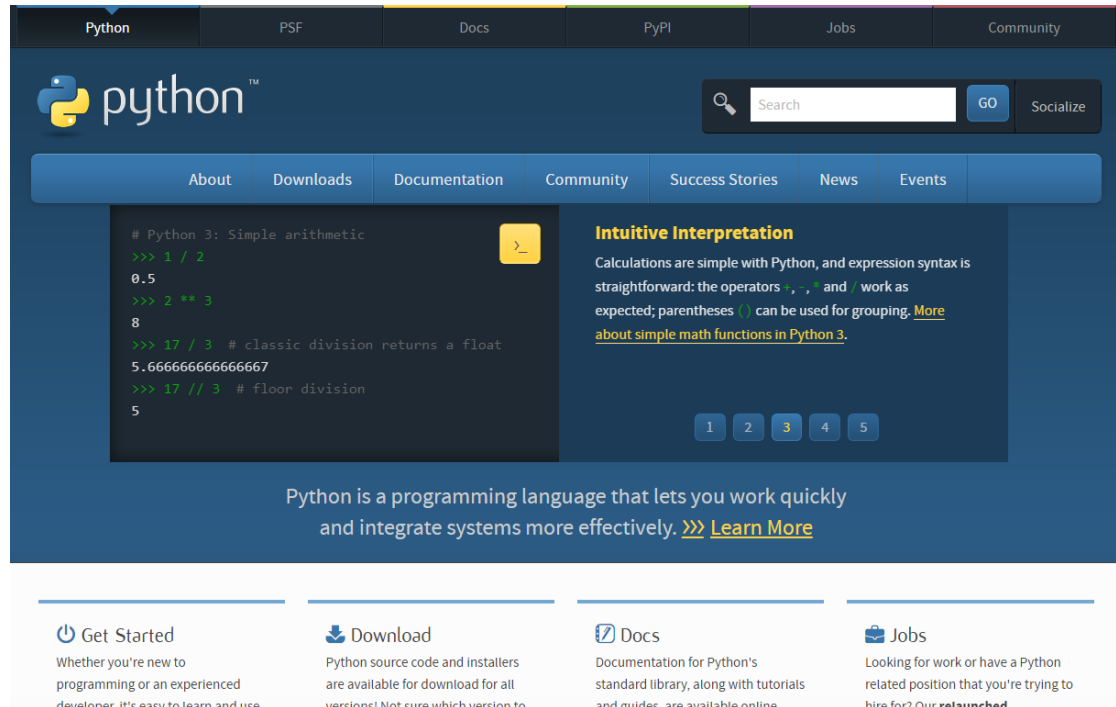




2. Hafta

PYTHON PROGRAMLAMA DİLİ NEDİR? NEDEN PYTHON? NASIL ÇALIŞIR?

Python



The screenshot shows the Python.org homepage with a dark blue header and navigation bar. The header includes the Python logo and the word "python" in a sans-serif font. To the right of the logo is a search bar with a magnifying glass icon and a "GO" button. Below the search bar is a "Socialize" button. The navigation bar contains links for "Python", "PSF", "Docs", "PyPI", "Jobs", and "Community". Below the navigation bar is a secondary navigation bar with links for "About", "Downloads", "Documentation", "Community", "Success Stories", "News", and "Events". The main content area features a code editor on the left with Python code for simple arithmetic, a yellow terminal icon, and a section titled "Intuitive Interpretation" on the right. The code editor shows the following code:

```
# Python 3: Simple arithmetic
>>> 1 / 2
0.5
>>> 2 ** 3
8
>>> 17 / 3 # classic division returns a float
5.666666666666667
>>> 17 // 3 # floor division
5
```

 The "Intuitive Interpretation" section explains that calculations are simple with Python and that the operators `+`, `-`, `*`, and `/` work as expected, while parentheses `()` can be used for grouping. It also includes a link to "More about simple math functions in Python 3." Below the main content area is a footer with four columns: "Get Started" (Whether you're new to programming or an experienced developer, it's easy to learn and use), "Download" (Python source code and installers are available for download for all versions! Not sure which version to), "Docs" (Documentation for Python's standard library, along with tutorials and guides, are available online), and "Jobs" (Looking for work or have a Python related position that you're trying to hire for? Our **relaunched**).

<https://www.python.org>



Python Tanıtım Kartı

- Geliştirici* ➤ Python Yazılım Vakfı
- İlk Tasarımcı* ➤ Guido van Rossum
- İlk çıkış tarihi* ➤ 20 Şubat 1991
- Lisansı* ➤ Python Yazılım Vakfı Lisansı (PSF)
- İşletim Sistemi* ➤ Çapraz platform (Win, Linux, Mac)
- Etkilendikleri* ➤ C, Haskel, Modula-3, Perl, Java, Lisp
- Etkiledikleri* ➤ D, Cobra, Falcon, Ruby, Go, JavaScript
- Güncel sürümü* ➤ **3.9.0**
- Dosya uzantıları* ➤ .py, .pyw, .pyc, .pyo, .pyd
- Web sitesi* ➤ www.python.org



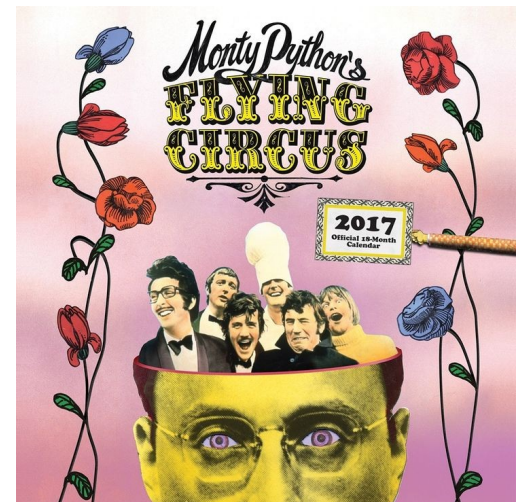
Python Adı? Okunuşu?

Python



- Python
- «Piton»
- «Paytın»
- «Pitın»
- Komedi grubu *Monty Python* 'a atıfla Guido tarafından Python olarak adlandırılmıştır.
- Monty Python'un Uçan Sirki (*Monty Python's Flying Circus*)
- İngiliz eskiz komedi dizisidir.
- Monty Python Komedi grubu.

<http://www.montypython.com>



Python Programlama Dili

- Yorumlamalı, etkileşimli, nesne yönelimli, yüksek seviyeli bir dildir.
- **Guido van Rossum** tarafından geliştirilmeye başlanmıştır.
- İlk olarak Şubat 1991 de (v0.9.0) ile başladığında Modula-3 den ödünç modül sistemi kullanılmıştır.



Guido van Rossum

- Hollandalı bir bilgisayar programcısıdır.
- Amsterdam Üniversitesi mezunu.
- Amerika Kaliforniya'da yaşıyor.
- Google'da çalıştı. (2005-2012)
- Halen Dropbox'da çalışıyor. (2013-ten günümüze)
- Kişisel sitesi: <https://gvanrossum.github.io/>



Guido / Python

- Guido van Rossum:
Welcome to Python Programming
- [https://
www.youtube.com/watch?v=SdH8fI1a
NZA](https://www.youtube.com/watch?v=SdH8fI1a_NZA)



- Guido Van Rossum :
Design of Computer Programs
- [https://
www.youtube.com/watch?v=bDo18m-
uOi8](https://www.youtube.com/watch?v=bDo18m-uOi8)

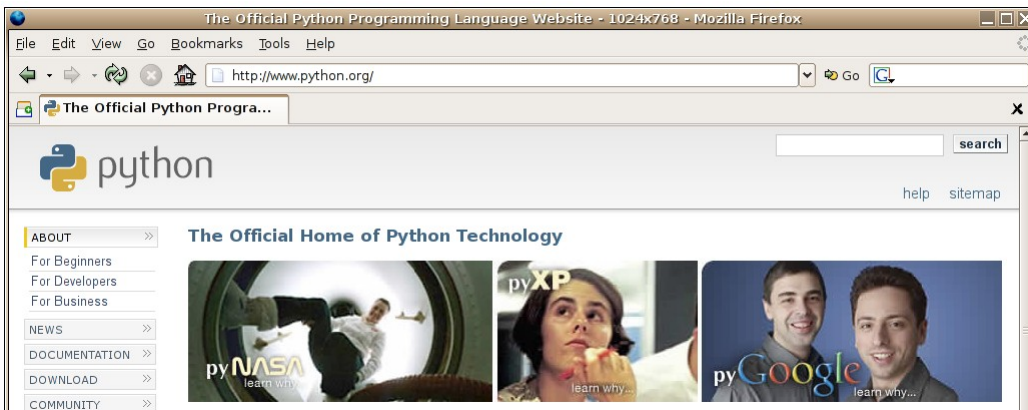
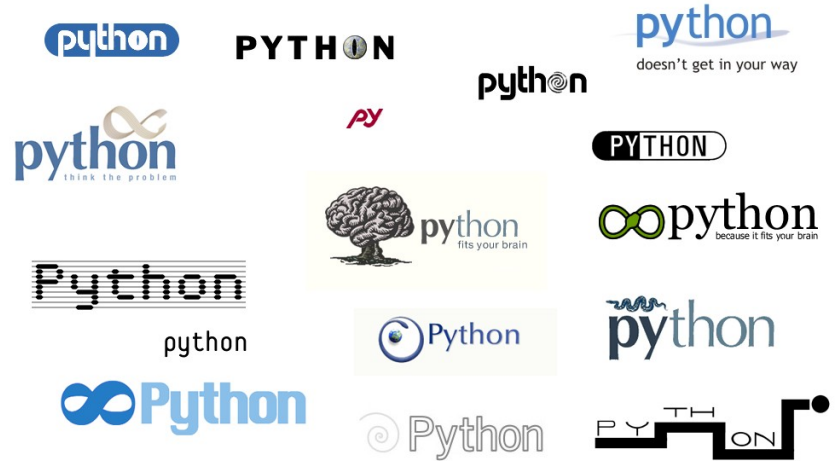
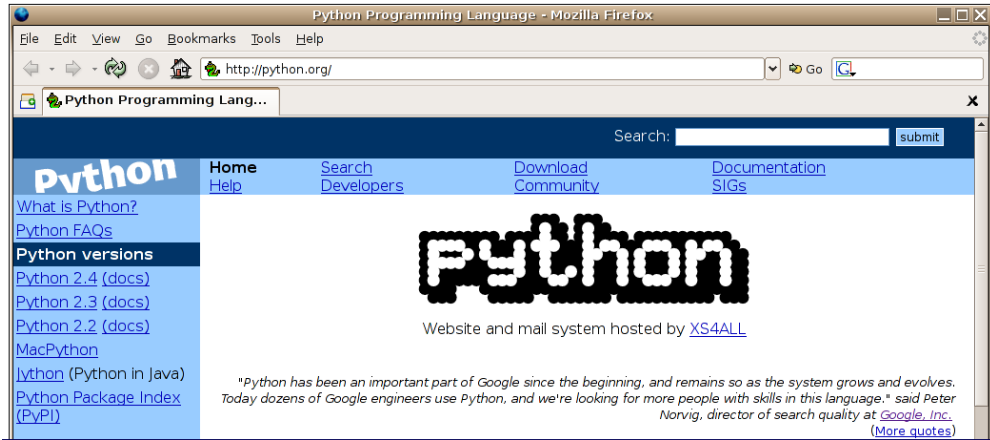


Neden Python? (Artıları)

- ✓ Net ve kolay okunabilen yazımı
- ✓ Güçlü ifade yeteneği
- ✓ Modüler yapısı
- ✓ Kolay anlaşılır nesne tabanlı programlama özellikleri
- ✓ Çalışma esnasında objelerin özelliklerini inceleyebilme imkanı
- ✓ Exception tabanlı hata yönetimi
- ✓ Çok yüksek seviye dinamik veri yapıları
- ✓ Çok geniş kütüphaneleri



Python Logo



Just von Rossum



<https://svn.python.org/www/branches/rest2web/beta.python.org/resources/docs/slideshow/s5.html>

Python Yazılım Vakfı (PSF)

- **Python Yazılım Vakfı** (Python Software Foundation, kısaca PSF)
6 Mart 2001 tarihinde Python programlama dilini korumak, geliştirmek ve uluslararası alanda gelişimine destek sağlamak amacıyla kurulmuş, kar amacı gütmeyen bir vakıftır.



<https://www.python.org/psf/>

Python Nereelerde?

Kullananlar

- Google
- Yahoo
- Dropbox
- Reddit
- NASA
- CERN
- Wikipedia
- ILM
- ITA
- ...

Gömülü Ürünler

- 3dsMax
- Blender
- Cinema 4d
- Lightware
- Maya
- FreeCAD
- Nuke
- GIMP
- ...

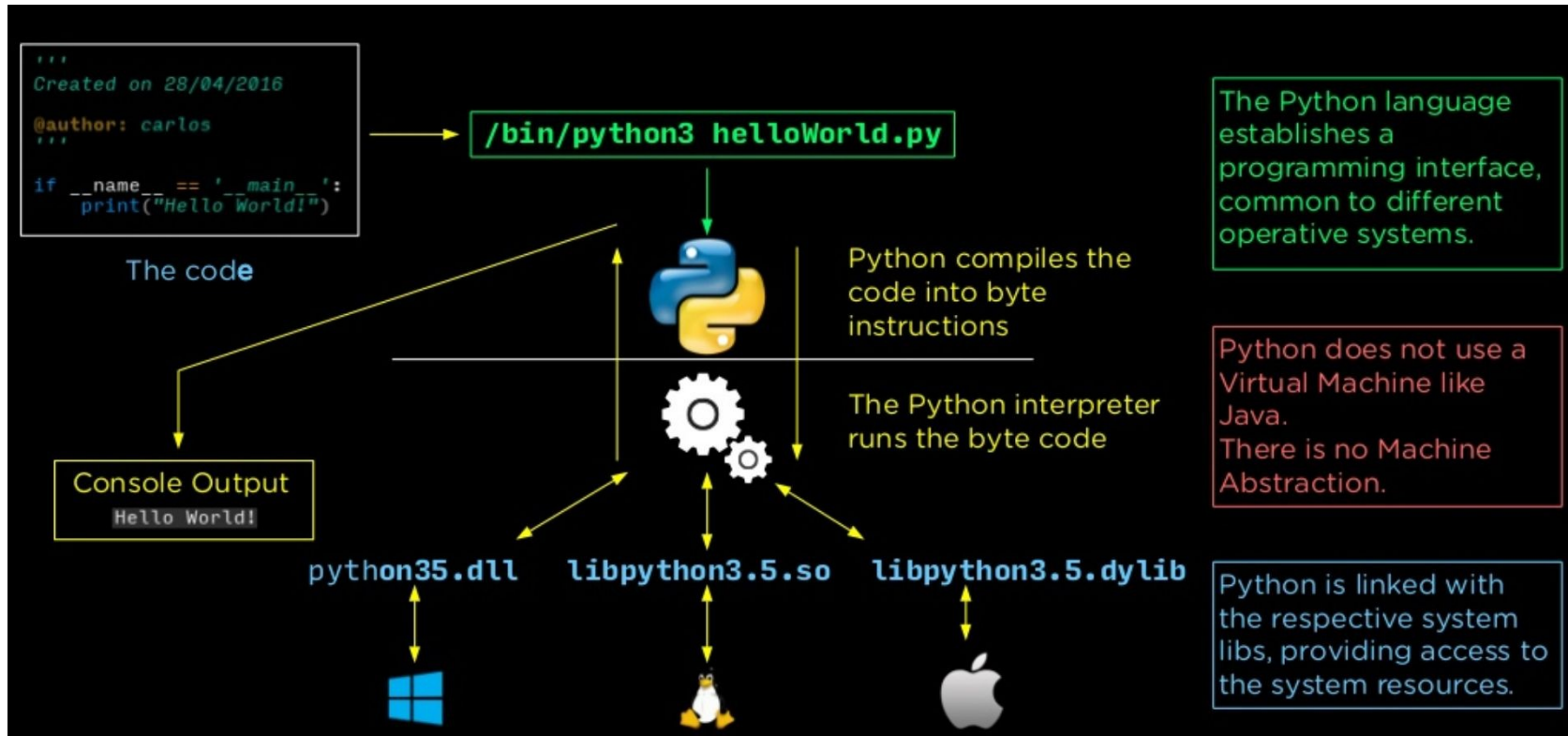
İşletim Sistemleri

- Linux
- FreeBSD
- MacOS
- OpenBSD
- AmigaOS
- Anaconda
(Ubuntu, RedHat, Fedora)
- Portage
(Gentoo paket yönetim sistemi)
- ...

LibreOffice, Yapay Zeka, Raspberry Pi,..

Nasıl Çalışır?

How The Python Interpreter Works



Neler Yapılır?

- Veri Analizi
- Bilimsel
- Sistem Yönetimi
- Web Uygulamaları
- Ağ ve Soket prog.
- Web Tarama (Örümcek)
- API oluşturma (REST)
- Machine Learning
- Yüz Tanıma
- Veri Madenciliği
- Parola Decrypt
- Oyun geliştirme
- Http Sunucu
- Yük Dengeleyici
- Plugin geliştirmek
- Modül ve Kütüphaneler

Ne Yapamazsınız?

- İşletim Sistemi yapamazsınız..

Güncel Sürümleri

- **Python 3.9.0** 5 Ekim 2020
- Python 3.8.6 24 Eylül 2020
- Python 3.7.9 17 Ağustos 2020
- Python 3.6.12 17 Ağustos 2020
- Python 3.5.10 5 Eylül 2020
-
- **Python 2.7.18** 2020-04-20
(Windows)
- 2009 dan itibaren 3.0 sürüm geliştirilmiş.
- v2 BİTTİ !
- **Python 2.7**
1 Ocak 2020 itibariyle v2 geliştirme sona erdi.

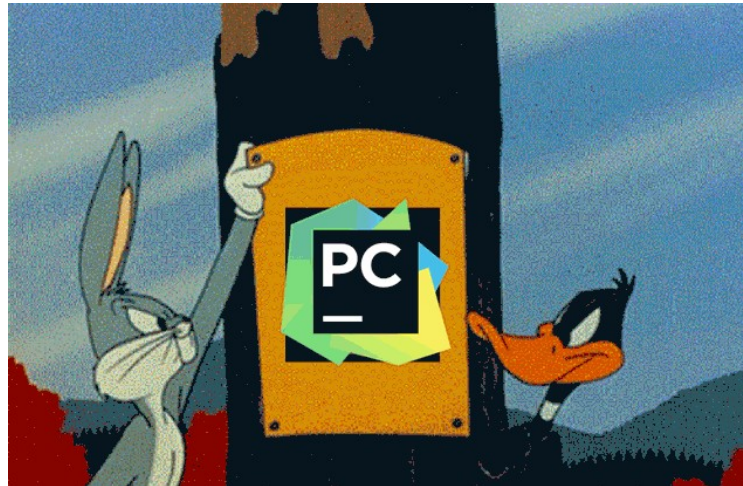
<https://www.python.org>

IDE'ler

Entegre Geliştirme Ortamı

(Integrated Development Environment)

- **IDLE**
- **Thonny**
- **PyCharm**
- VS Code
- Spyder
- Komodo
- LiClipse
- NetBeans
- SubLime
- Wing
- PyStudio
- PyScripter
- Idlex
- Pyzo
- ...



Name	Platform	Updated
 Thonny	Windows, Linux, Mac OS X, more	2020
Komodo	Windows/Linux/Mac OS X	2017
 LiClipse	Linux/Mac OS X/Windows	2018
NetBeans	Linux, Mac, Solaris, Windows	2016
 PyCharm	Linux/Mac OS X/Windows	2018
 Python for VS Code	Linux/Mac OS X/Windows	2018
 KDevelop	Linux/Mac OS X/(Windows)	2017
PyDev	Eclipse	2018
 Wing	Windows, Linux, Mac OS X	2020
 PyScripter	Windows	2012
 Spyder	Windows/Linux/macOS	2018
IDLE	Windows/Linux/Mac OS X/All Tk Platforms	2018
 Idlex	Windows/Linux/Mac OS X/All Tk Platforms	2016
 u.dev	Windows (needs to be compiled manually for other platforms)	2010
 Pyzo (formerly IEP)	Windows/Linux/Mac OS X	2018
 PythonToolKit (PTK)	Windows/Linux/Mac OS X	2014
 PyStudio	Windows/Linux/Mac OS X	2012
 Python Tools for Visual Studio	Windows	2017
 Exedore	Mac OS X	2015

Kaynak: <https://wiki.python.org/moin/IntegratedDevelopmentEnvironments>

IDLE



- Python **IDLE**
- **I**ntegrated **D**evelopment and **L**earning **E**nvironment
(Integrated **D**evelopment **E**nvironment)
- Entegre Geliştirme ve Öğrenme Ortamı
(Entegre Geliştirme Ortamı)
- Python için entegre bir geliştirme ortamıdır. (Etkileşimli Kabuk - interactive shell)

```
Python 3.6.4 Shell
File Edit Shell Debug Options Window Help
Python 3.6.4 (v3.6.4:d48eceb, Dec 19 2017, 06:04:45) [MSC v.1900
32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more information.
>>> |
```

```
Python 3.8.0 (default, Nov 14 2019, 22:29:45)
[GCC 5.4.0 20160609] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> |
```

```
Python 3.7.3 (default, Jul 25 2020, 13:03:44)
[GCC 8.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> |
```

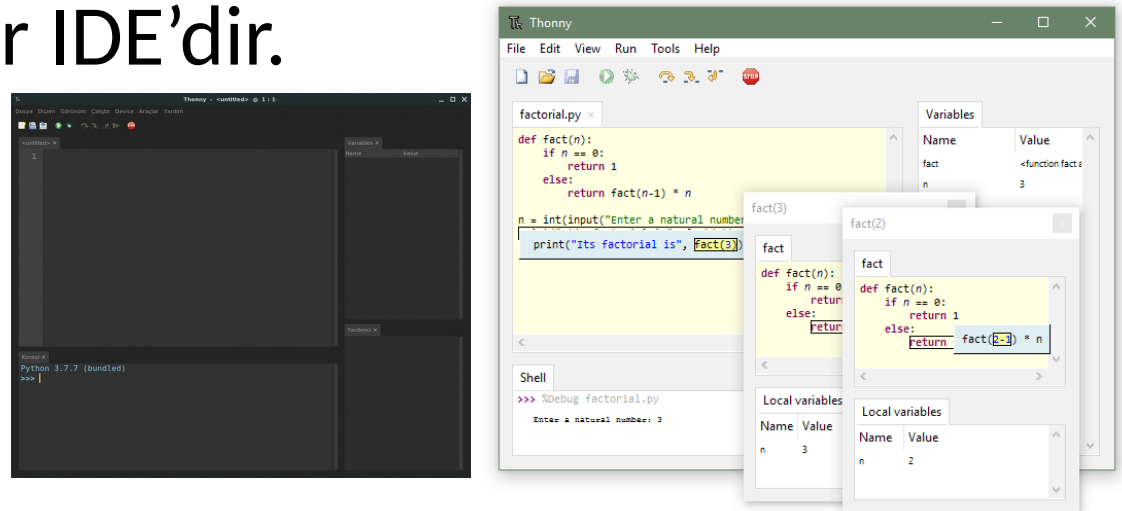
<https://www.python.org>

Thonny



- Yeni başlayanlar için bir Python IDE'dir.
- Adım adım ifade değerlendirmesi ve değişken değeri gibi bildirimleri ile kolay kullanımlıdır.
- Tartu Üniversitesi (Estonya) tarafından MIT lisansı ile dağıtılan Windows, Linux, MacOS'da kullanılabilen bir IDE'dir.

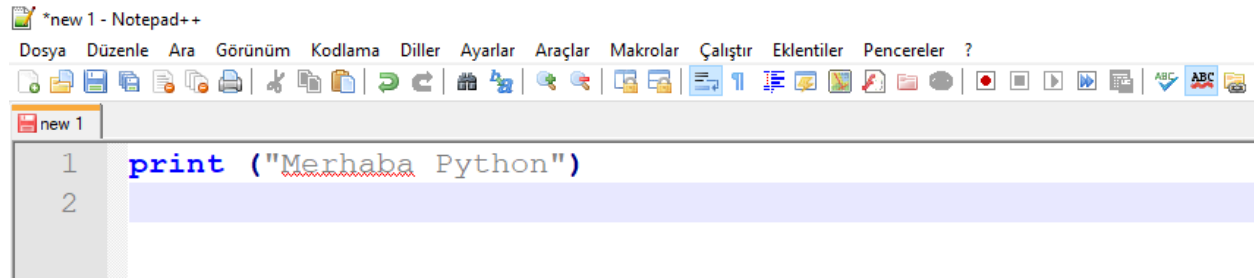
<http://thonny.org>



Notepad++



- GPL ile dağıtılan açık kaynak kodlu bir kaynak kod düzenleyicisidir. *Scintilla* metin editörü temel alınarak geliştirilmiştir.
- Türkçe dahil **84** ayrı konuşma diline destek veren bir arabirimi vardır.
- **73** ayrı programlama dilinin söz dizimi ve renklendirmesini yapabilen Windows ortamının popüler editörüdür.



<https://notepad-plus-plus.org/>



PyCharm



- JetBrains firması tarafından ticari bir ürün olarak özellikle python dili için geliştirilen bir IDE'dir.
- Kod analizi, grafiksel bir hata ayıklayıcı, entegre bir ünite test cihazı, sürüm kontrol sistemleri ile entegrasyon (VCSes) ve Django ile web geliştirme desteği sunar.
- Windows, Linux ve MacOS da kullanılabilen Community(Topluluk) ve Ticari lisanslı iki sürümü vardır.

<https://www.jetbrains.com/pycharm/>





PC **PyCharm**

Python IDE
for Professional Developers



```

djtp_first_steps > polls > tests.py
tests.py x
20
21
22 response = self.client.get(reverse('polls:index'))
23 self.assertEqual(response.status_code, 200)
24 self.assertContains(response, "No polls are available.")
25 self.assertQuerysetEqual(response.context['latest_question_list'], [])
26 self.test
27
28 def test_index_view_with_a_future_question(self) QuestionViewTests
29
30 def test_index_view_with_a_past_question(self) QuestionViewTests
31
32 def test_index_view_with_future_question_and_past_question QuestionVi...
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34 def test_index_view_with_no_questions(self) QuestionViewTests
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36 def test_index_view_with_two_past_questions(self) QuestionViewTests
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Online Python Editörleri

- <https://www.python.org/shell>
- <https://python.codnex.net/>
- https://www.onlinegdb.com/online_python_compiler
- <http://pythonfiddle.com/>
- <https://repl.it/languages/python3>
- http://www.compileonline.com/execute_python_online.php
- <https://www.remoteinterview.io/online-python-compiler>
- https://www.tutorialspoint.com/execute_python_online.php
- <http://www.pythontutor.com/visualize.html#mode=edit>
- <https://www.jdoodle.com/python-programming-online>
- <http://techmums.co/python.html>
- http://rextester.com/l/python3_online_compiler
- <http://www.muratkara.com/python/shell/>

(çalışması için öncelikle python.org/shell girilmelidir.)



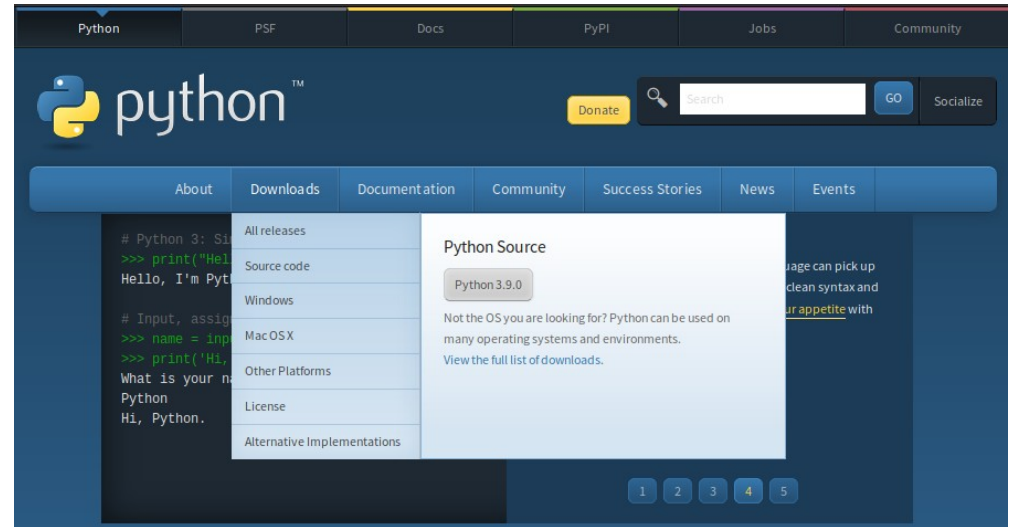
Haftaya Hazırlık (İndiriniz)

- **Python (3.9.0)**

- www.python.org

python-3.9.0.exe ~27MB

Python-3.9.0.tar.xz

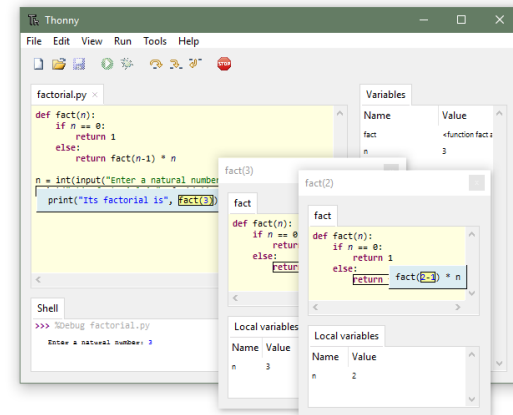
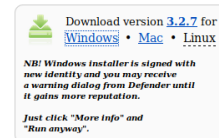


- **Thonny**

- <http://thonny.org/>

thonny-3.2.7.exe ~15MB

Thonny
Python IDE for beginners



>>> Kurulum ve giriş haftaya..



3. Hafta

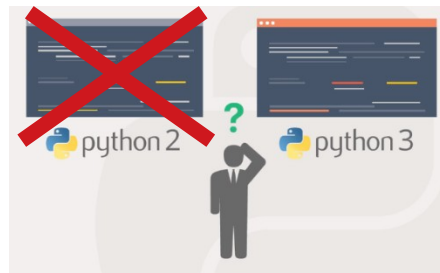
PYTHON 2X VE 3X SÜRÜMLERİ, PYTHON KURULUMU

Sürümleri

- Güncel Sürümler;
 - **Python 3.9.0**
 - Python 2.7.18



- Python 1991 den beri gelişimini devam ettiren bir dildir.
- Gelişim yeni sürümler ile devam etmektedir.
- Python3, python2 ye göre çok daha güçlüdür.
- Python2, 1 Ocak 2020 de desteği kesilmiş bir sürümdür.



Python Sürümleri

Python Sürümleri

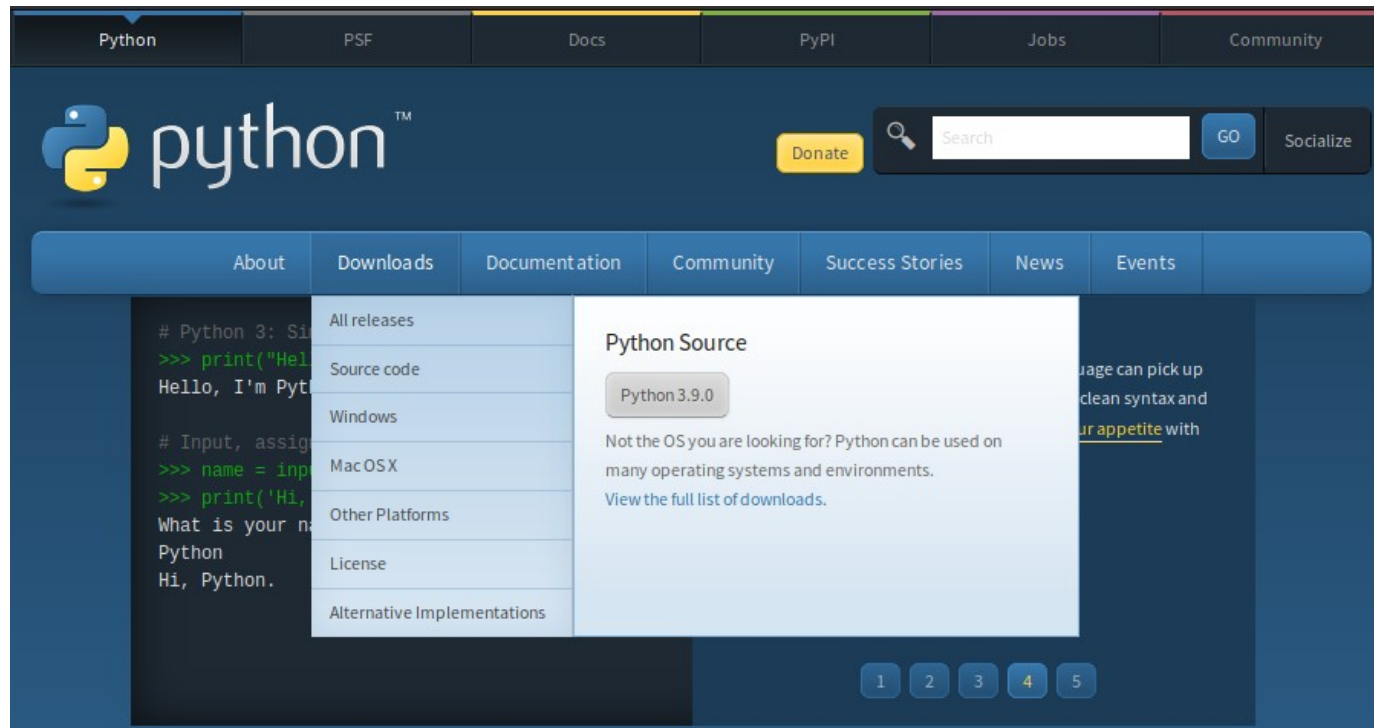


Örneğin;
2.7.11 > 2.x
serisine aittir.

Örneğin;
3.5.1 > 3.x
serisine aittir.

Kurulumlar

- **Python (3.9.0)**
 - www.python.org
- python-3.9.0.exe ~27MB



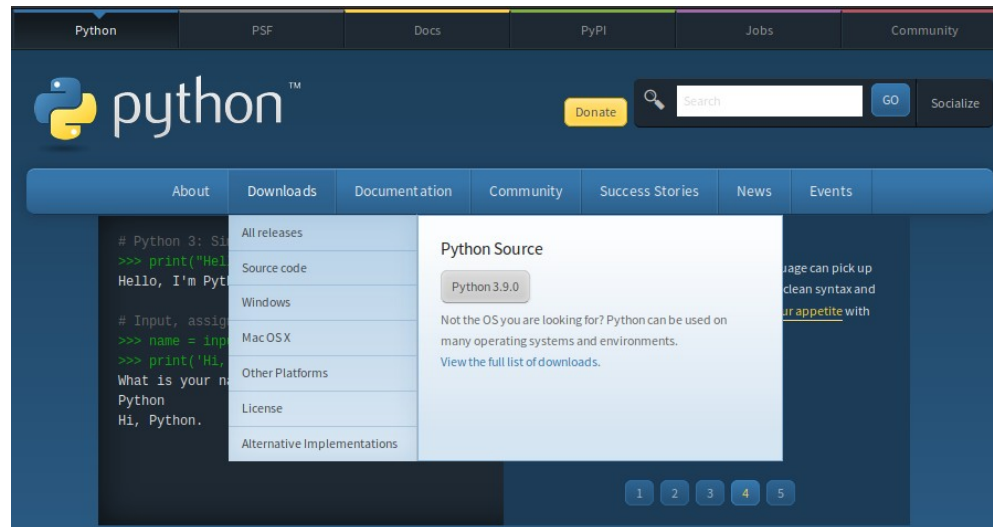
Kurulumlar

- **Python (3.x)**
 - www.python.org

İlgili işletim sistemine uygun sürüm indirilir..

Kurulumlar;

- Windows
- MacOS
- Linux



Windows

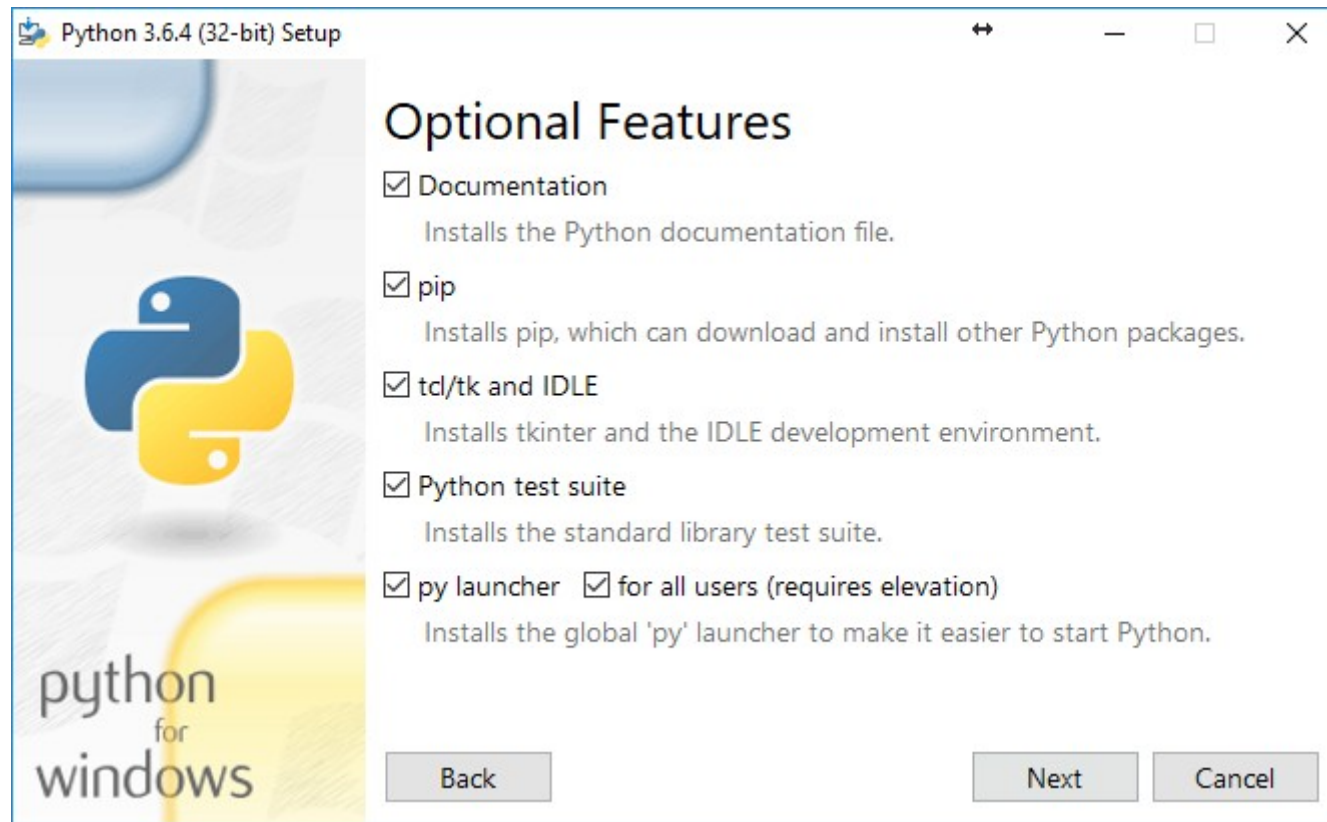
- İndirmiş olduğumuz dosyaya çift tıklayıp başlayabiliriz.



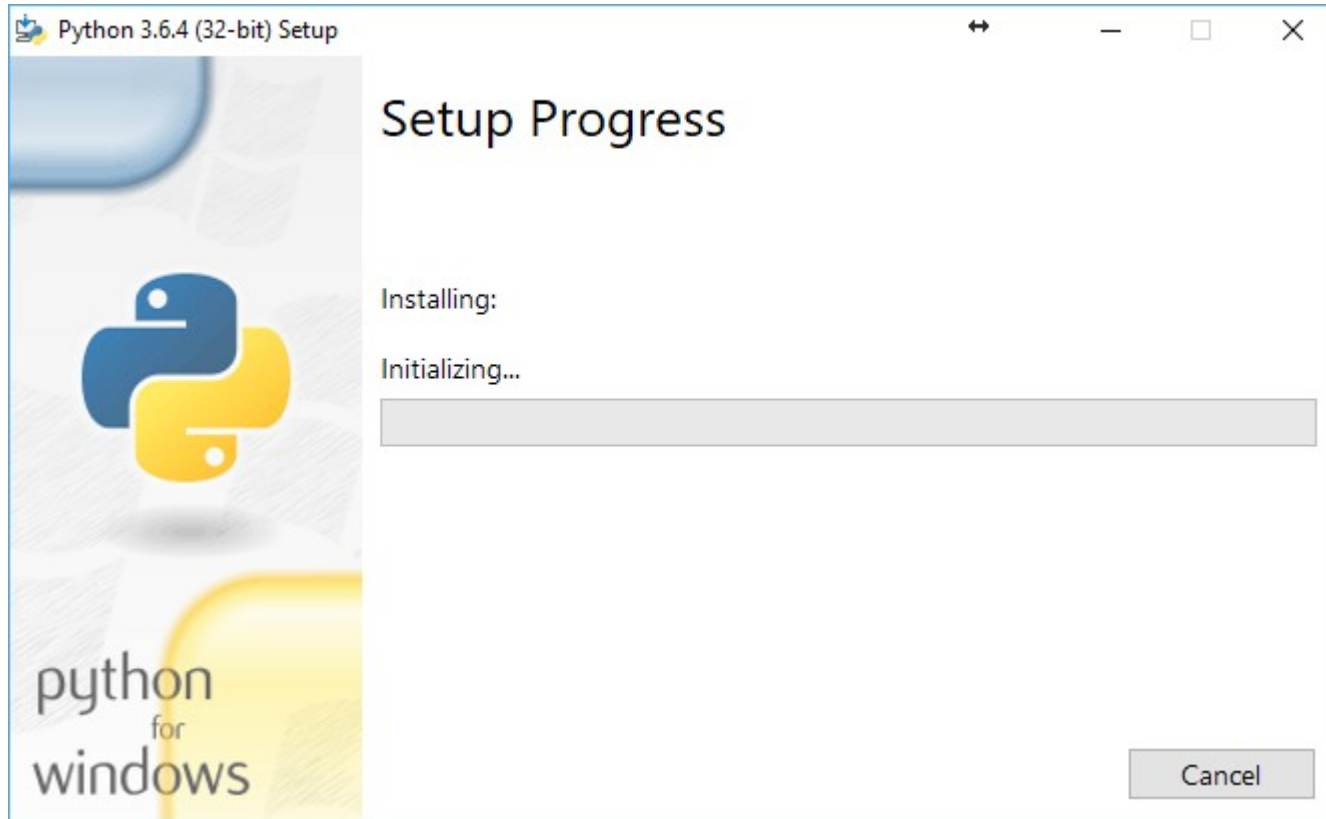
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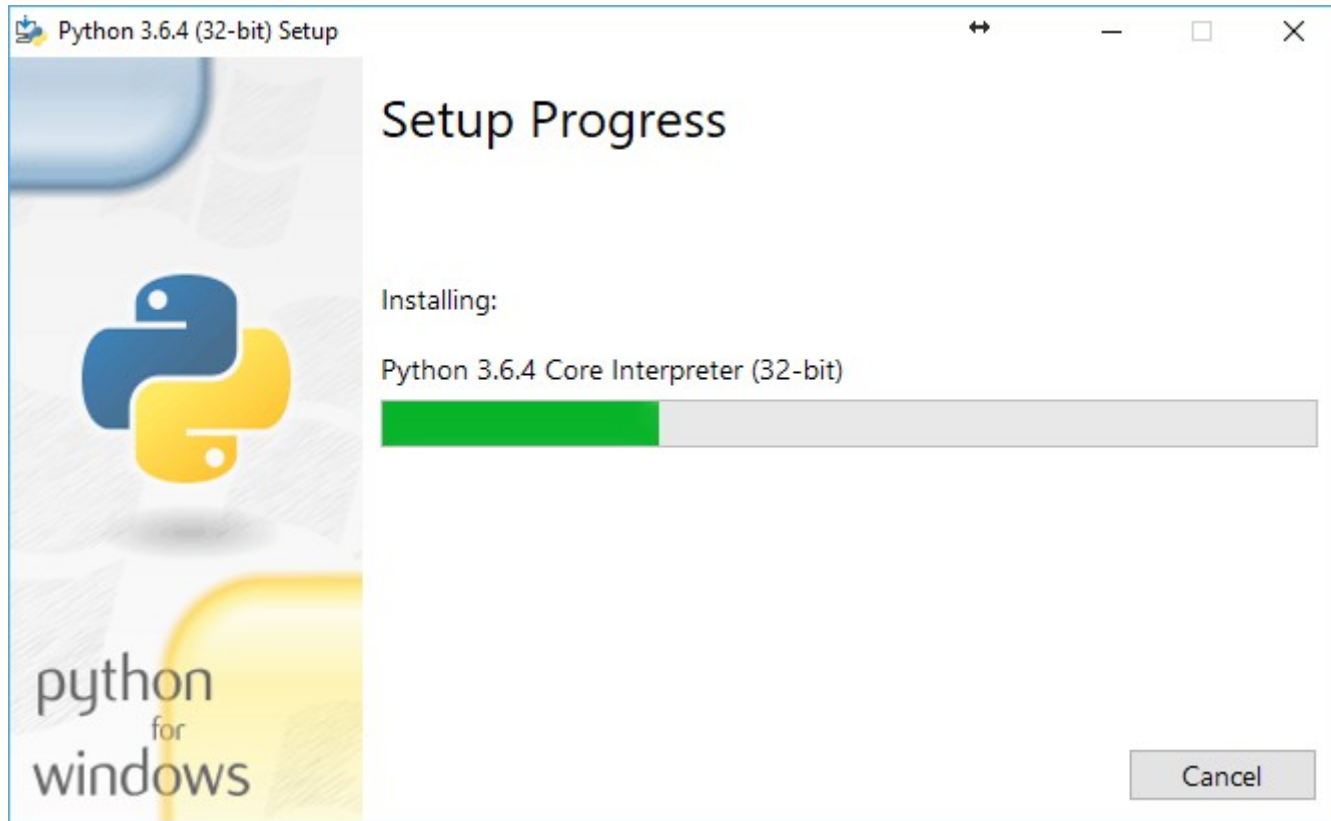
Windows



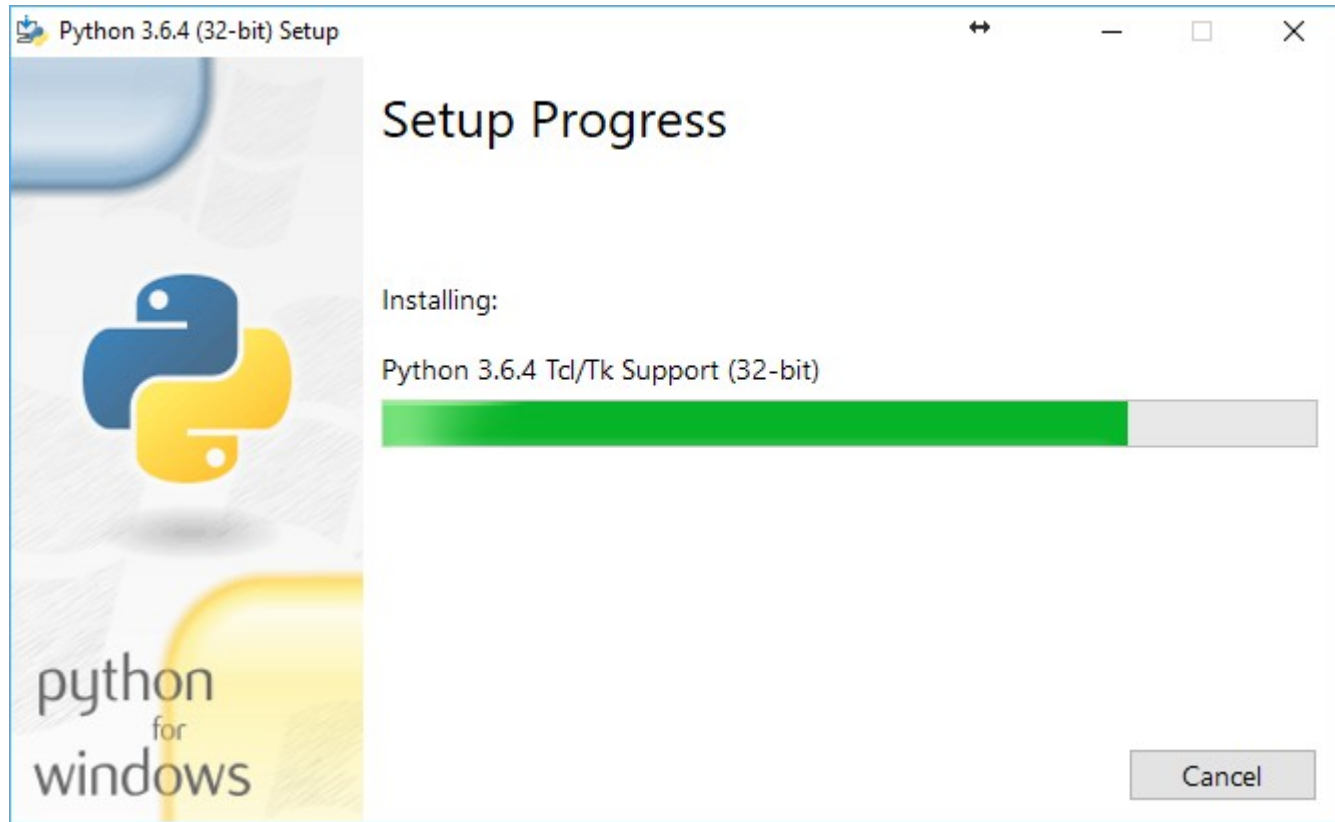
Windows



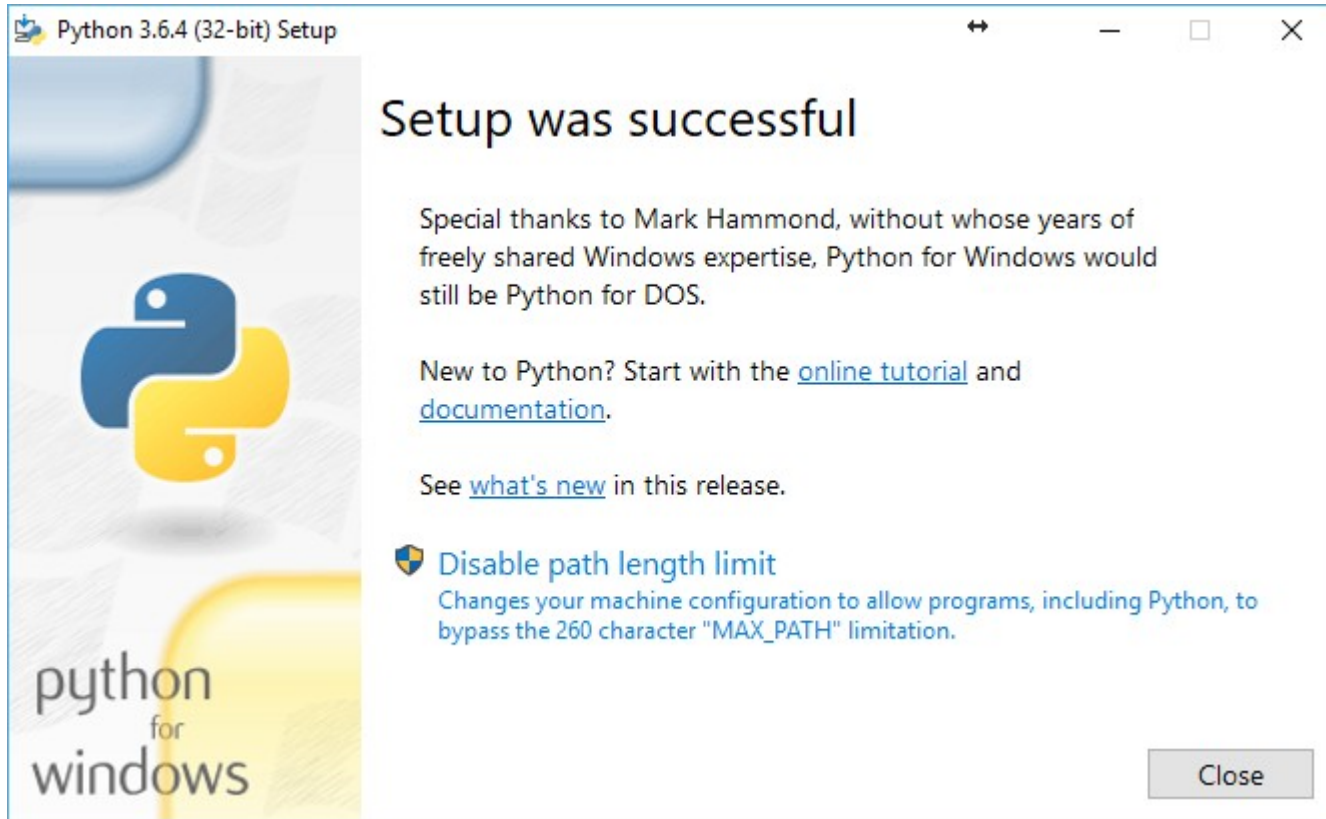
Windows



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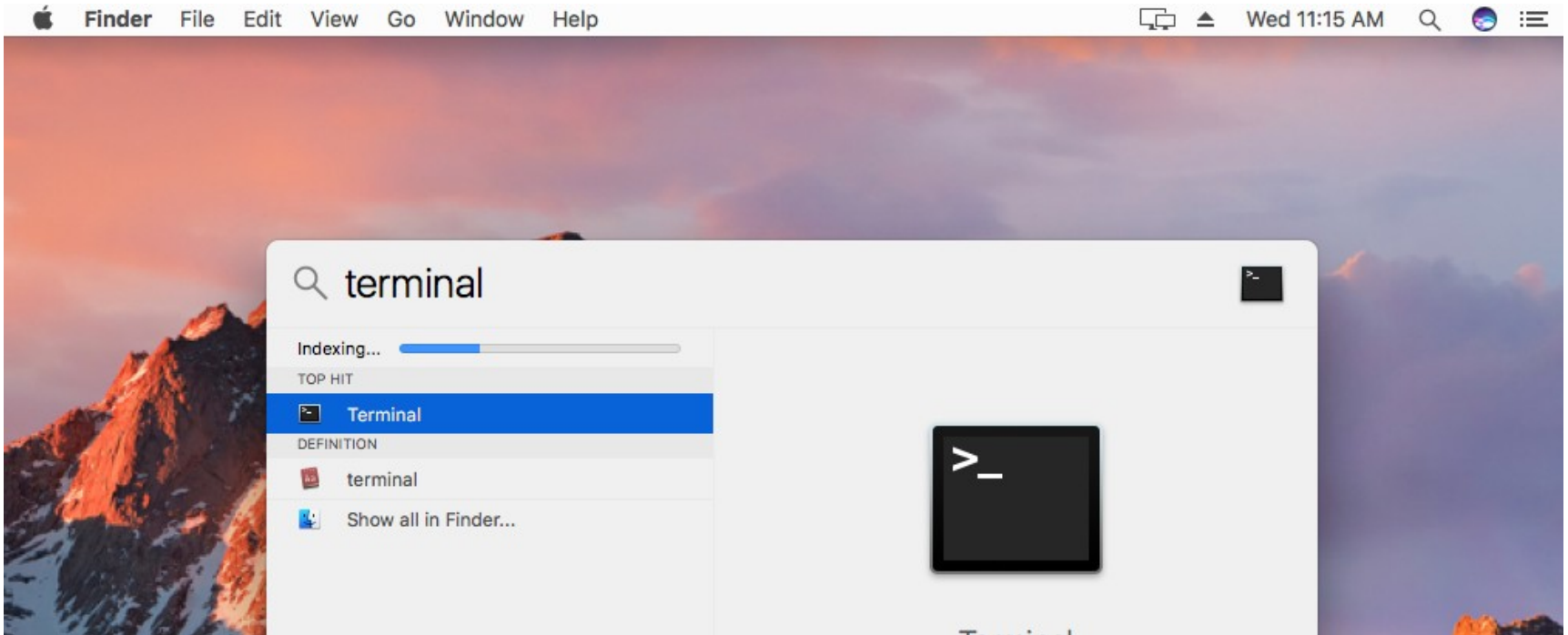
Windows



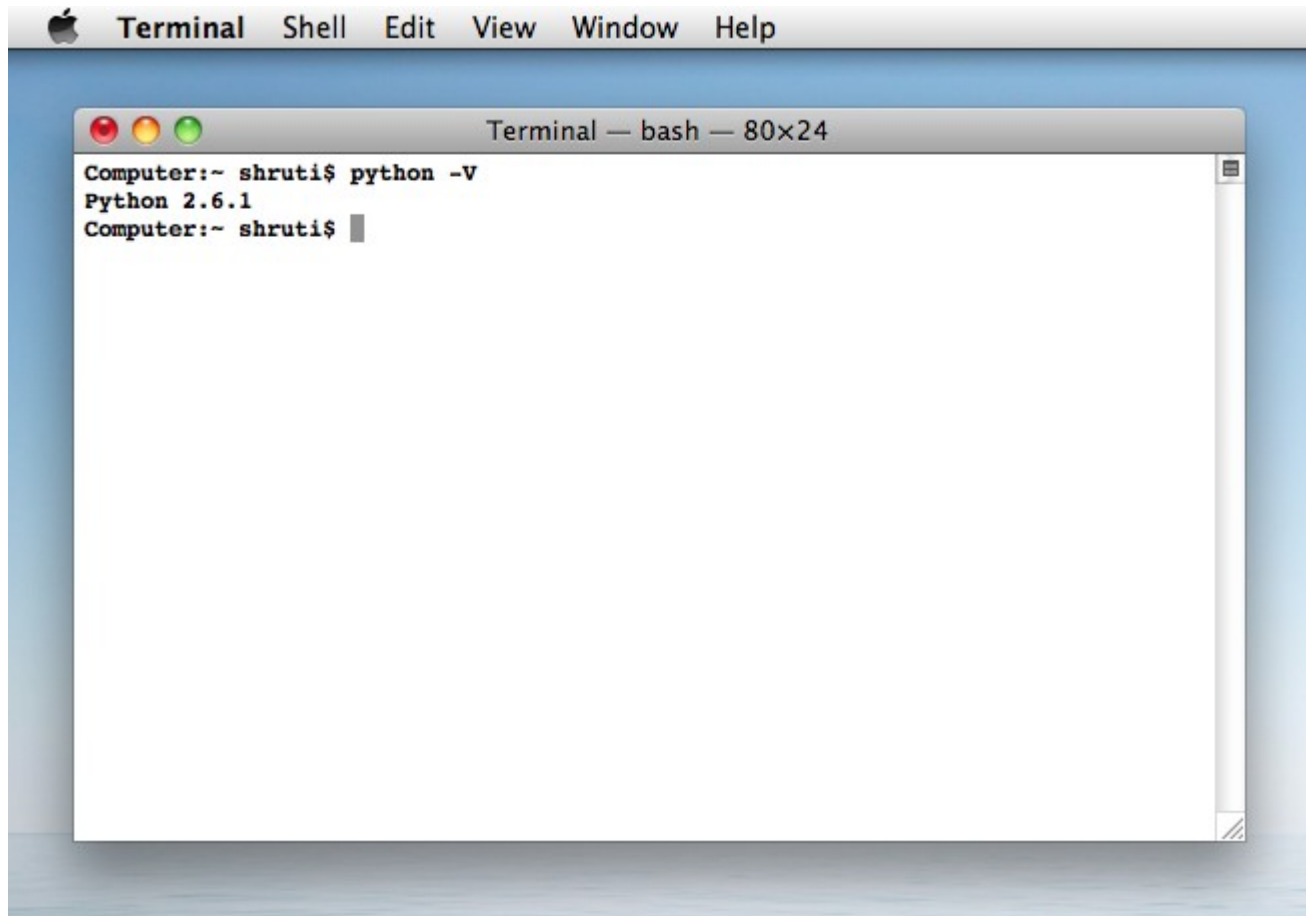
Windows

A screenshot of a 'Python 3.6.4 Shell' window. The title bar shows the application name and standard window controls. The menu bar includes 'File', 'Edit', 'Shell', 'Debug', 'Options', 'Window', and 'Help'. The main text area displays the Python version and build information: 'Python 3.6.4 (v3.6.4:d48eceb, Dec 19 2017, 06:04:45) [MSC v.1900 32 bit (Intel)] on win32'. It also shows the prompt 'Type "copyright", "credits" or "license()" for more information.' and the interactive prompt '>>> |'.A screenshot of a 'Komut İstemi - python' (Command Prompt - python) window. The title bar shows the application name and standard window controls. The main text area shows the command 'C:\>python' being executed, followed by the same Python version and build information as the shell window: 'Python 3.6.4 (v3.6.4:d48eceb, Dec 19 2017, 06:04:45) [MSC v.1900 32 bit (Intel)] on win32'. It also shows the prompt 'Type "help", "copyright", "credits" or "license" for more information.' and the interactive prompt '>>>'.

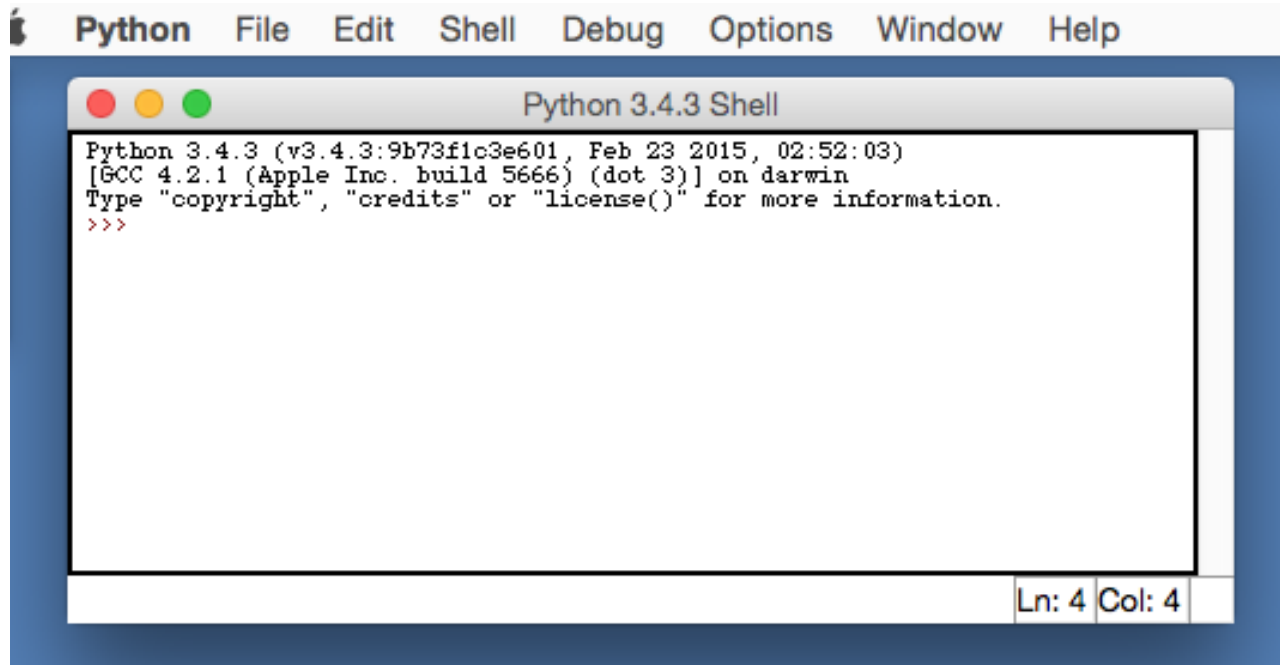
MacOS



MacOS



MacOS



MacOS

- Homebrew paket kurulum yöneticisidir.

\$ brew search python

\$ brew update

\$ brew install python3

\$ brew upgrade python3

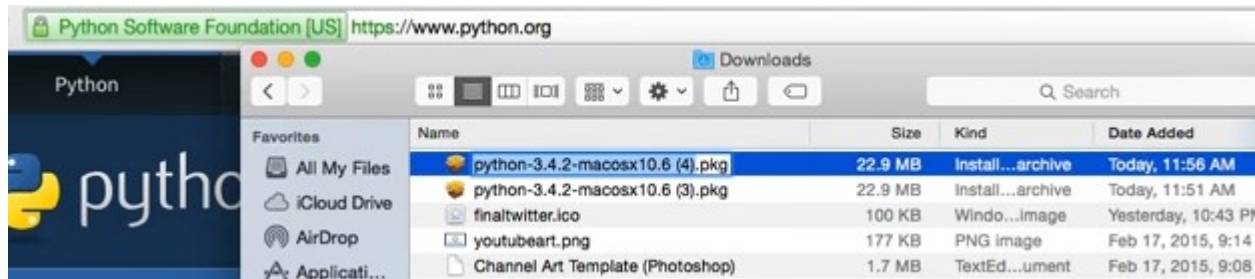
MacOS



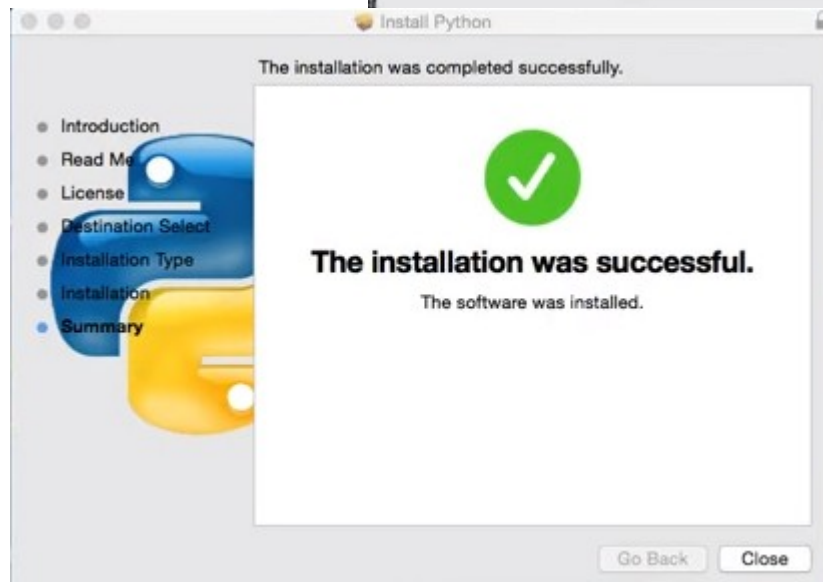
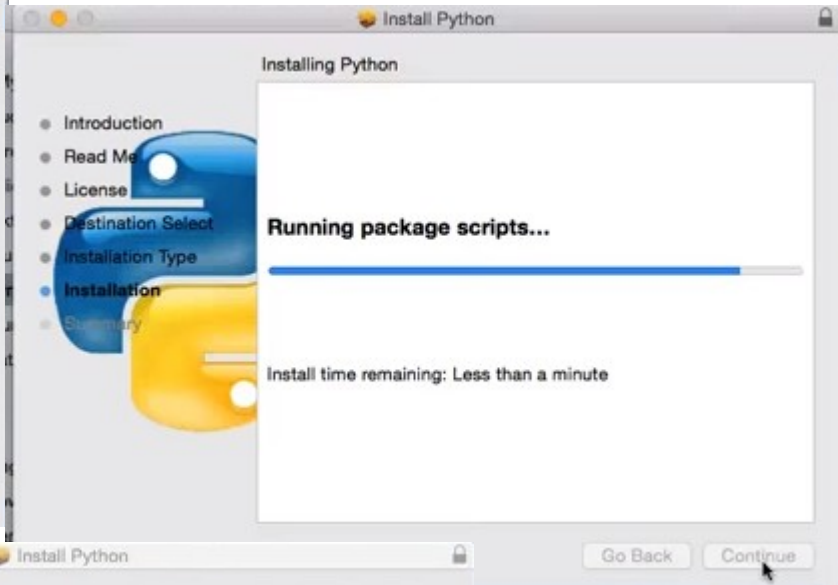
MacOS

Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		e19e75ec81dd04de27797bf3f9d918fd	26724009	SIG
XZ compressed source tarball	Source release		6ebfe157f6e88d9eabfbaf3fa92129f6	18866140	SIG
macOS 64-bit installer	Mac OS X	for OS X 10.9 and later	16ca86fa3467e75bade26b8a9703c27f	31132316	SIG
Windows help file	Windows		9ea6fc676f0fa3b95af3c5b3400120d6	8757017	SIG
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64T/x64	60d0d94337ef657c2cca1d3d9a6dd94b	8387074	SIG
Windows x86-64 executable installer	Windows	for AMD64/EM64T/x64	b61a33dc28f13b561452f3089c87eb63	28158664	SIG
Windows x86-64 web-based installer	Windows	for AMD64/EM64T/x64	733df85afb160482c5636ca09b89c4c8	1364352	SIG
Windows x86 embeddable zip file	Windows		d81fc534080e10bb4172ad7ae3da5247	7553872	SIG
Windows x86 executable installer	Windows		4a2812db8ab9f2e522c96c7728cfcccb	27066912	SIG
Windows x86 web-based installer	Windows		cdbfa799e6760c13d06d0c2374110aa3	1327384	SIG

MacOS



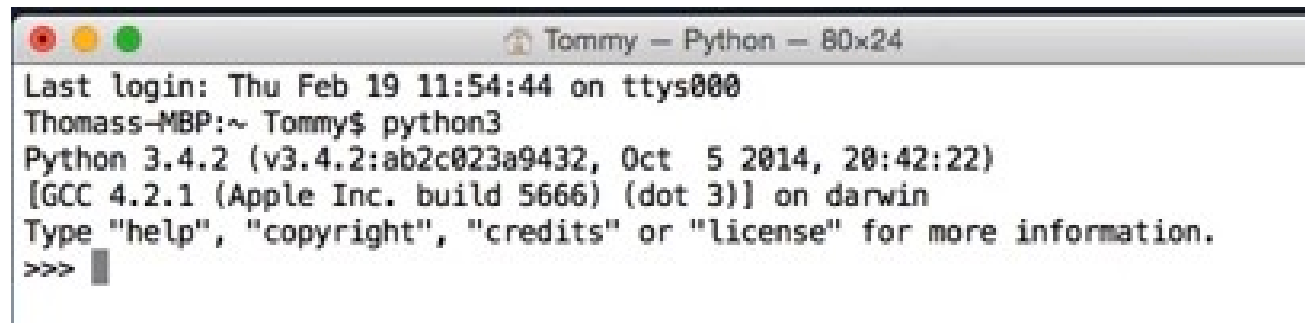
MacOS



MacOS

- Python3 açıyoruz..
- Terminal

\$ python3

A screenshot of a macOS Terminal window. The title bar at the top shows three colored window control buttons (red, yellow, green) on the left and the text 'Tommy - Python - 80x24' on the right. The terminal content displays the output of running 'python3'. It shows the last login time as 'Thu Feb 19 11:54:44 on ttys000', the user 'Thomass-MBP:~ Tommy', and the Python version 'Python 3.4.2 (v3.4.2:ab2c023a9432, Oct 5 2014, 20:42:22)'. It also mentions the compiler '[GCC 4.2.1 (Apple Inc. build 5666) (dot 3)] on darwin' and provides instructions to type 'help', 'copyright', 'credits', or 'license' for more information. The prompt '>>>' is visible at the bottom with a cursor.

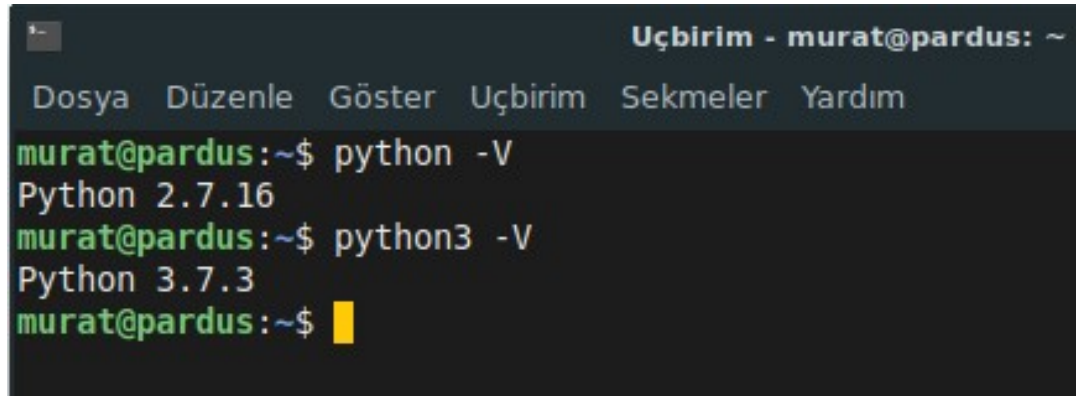
Linux

- Hemen hemen bütün Linux sistemlerinde kurulu geliyor.
- Sürüm bilgisini öğrenmek için;

\$ python -V

veya

\$ python3 -V

A terminal window titled "Uçbirim - murat@pardus: ~" with a menu bar containing "Dosya", "Düzenle", "Göster", "Uçbirim", "Sekmeler", and "Yardım". The terminal shows two commands and their outputs: first, "murat@pardus:~\$ python -V" followed by "Python 2.7.16"; second, "murat@pardus:~\$ python3 -V" followed by "Python 3.7.3". The prompt "murat@pardus:~\$" is followed by a yellow cursor bar.

```
Uçbirim - murat@pardus: ~
Dosya  Düzenle  Göster  Uçbirim  Sekmeler  Yardım
murat@pardus:~$ python -V
Python 2.7.16
murat@pardus:~$ python3 -V
Python 3.7.3
murat@pardus:~$
```

Linux

Debian tabanlı sistemlerde paket ile kurmak için;

```
$ sudo apt-get update
```

```
$ sudo apt-get upgrade -y
```

```
$ sudo apt-get dist-upgrade
```

```
$ sudo apt-get install python3.9
```

veya

```
$ sudo apt-get install python3-all-dev
```

Linux

Kaynak kod ile kurmak için;

```
#wget https://www.python.org/ftp/python/3.9.0/Python-3.9.0.tar.xz
```

```
#tar xJf Python-3.9.0.tar.xz
```

```
# cd Python-3.9.0
```

```
# ./configure
```

```
# make
```

```
# make install
```

Linux

CentOs/RedHat sistemlerine kurmak için;

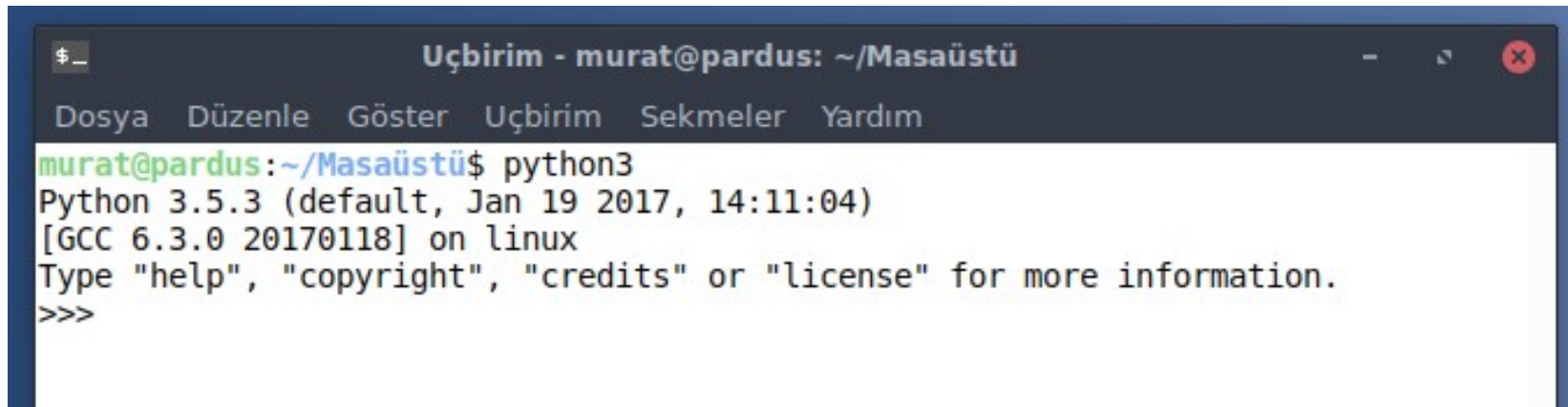
```
# yum -y groupinstall development
```

```
# yum -y install zlib-devel
```

```
# yum -y install python39u
```


Linux

- Python3 açıyoruz..
\$ python3

A terminal window titled "Uçbirim - murat@pardus: ~/Masaüstü" with standard window controls. The menu bar includes "Dosya", "Düzenle", "Göster", "Uçbirim", "Sekmeler", and "Yardım". The terminal shows the command "python3" being executed, followed by the output: "Python 3.5.3 (default, Jan 19 2017, 14:11:04)", "[GCC 6.3.0 20170118] on linux", and a prompt to type "help", "copyright", "credits" or "license" for more information. The prompt ">>>" is visible at the end of the output.

```
$ _  
Uçbirim - murat@pardus: ~/Masaüstü  
Dosya  Düzenle  Göster  Uçbirim  Sekmeler  Yardım  
murat@pardus:~/Masaüstü$ python3  
Python 3.5.3 (default, Jan 19 2017, 14:11:04)  
[GCC 6.3.0 20170118] on linux  
Type "help", "copyright", "credits" or "license" for more information.  
>>>
```

Başlıyoruz..

- Kurulumlar tamam.. Python ı açıyoruz..



```
print("Hello, world!")
```

IDLE'ı açıyoruz..

- Windows Python IDLE

A screenshot of the Python 3.6.4 Shell window. The window has a title bar with the text "Python 3.6.4 Shell" and standard window controls (minimize, maximize, close). Below the title bar is a menu bar with the options "File", "Edit", "Shell", "Debug", "Options", "Window", and "Help". The main area of the window is a text editor showing the following text: "Python 3.6.4 (v3.6.4:d48eceb, Dec 19 2017, 06:04:45) [MSC v.1900 32 bit (Intel)] on win32", "Type \"copyright\", \"credits\" or \"license()\" for more information.", and a prompt ">>> |". The status bar at the bottom right of the window shows "Ln: 3 Col: 4".

```
Python 3.6.4 (v3.6.4:d48eceb, Dec 19 2017, 06:04:45) [MSC v.1900
32 bit (Intel)] on win32
Type "copyright", "credits" or "license()" for more information.
>>> |
```

İlk Programımız

```
>>> print ("Merhaba Dünya")
```

Merhaba Dünya

Python 3.6.4 Shell

File Edit Shell Debug Options Window Help

Python 3.6.4 (v3.6.4:d48eceb, Dec 19 2017, 06:04:45) [MSC v.1900 32 bit (Intel)]
on win32

Type "copyright", "credits" or "license()" for more information.

```
>>> print ("Merhaba Dünya")
```

Merhaba Dünya

```
>>> |
```